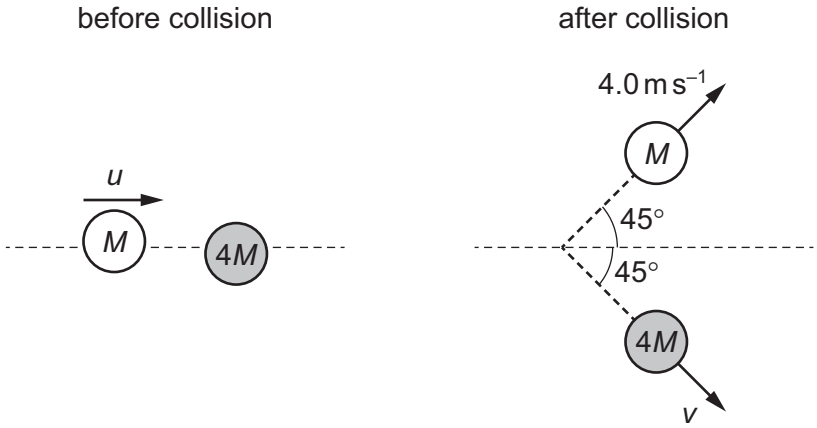
12 A disc of mass M is moving across a horizontal frictionless surface with constant velocity u. It collides with a stationary disc of mass $4M$.

The diagram shows the view from above of the motion of the two discs before and after the collision.



What is the initial velocity u of the disc of mass M?

A $1.1\,m\,s^{-1}$ **B** $1.4\,m\,s^{-1}$ **C** $3.5\,m\,s^{-1}$ **D** $5.7\,m\,s^{-1}$